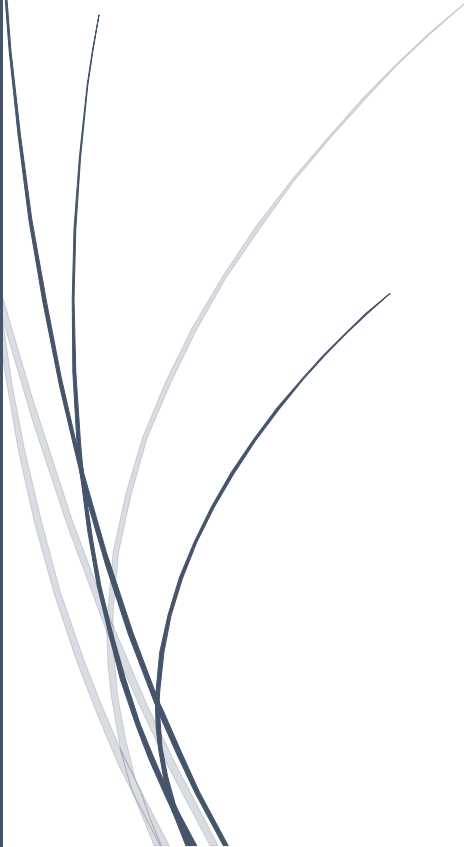


A dark blue vertical bar on the left side of the page. A blue arrow points to the right from this bar, containing the date.

5/12/2026

Practical Cybersecurity Laboratory Report

BIS 303 Access Control &
Penetration test

Several thin, curved lines in dark blue and light grey that sweep upwards from the bottom left towards the center of the page.

Jacques RENANAGA
SIIMT UNIVERSITY COLLEGE

USING NMAP ON WINDOWS AND LINUX WITH

1. Introduction

This practical focused on performing network reconnaissance and vulnerability assessment using Kali Linux and Nmap against the Metasploitable2 vulnerable machine.

The objective of the practical was to identify:

- Active hosts
- Open ports
- Running services
- Service versions
- Operating system information
- Potential security weaknesses

Metasploitable2 is an intentionally vulnerable Linux machine designed for cybersecurity training and penetration testing practice.

2. Objectives

The objectives of this practical were:

- To understand network scanning
- To use Nmap for reconnaissance
- To identify open ports and services
- To detect operating system information

- To analyze vulnerable services
- To understand basic penetration testing methodology

3. Tools and Environment

Tool	Purpose
Kali Linux	Attacker machine
Metasploitable2	Vulnerable target machine
Nmap	Network scanning tool
VMware	Virtualization platform

4. Network Configuration

Device	IP Address
Kali Linux	192.168.126.128
Metasploitable2	192.168.126.129

Both virtual machines were connected on the same virtual network.

5. Connectivity Test

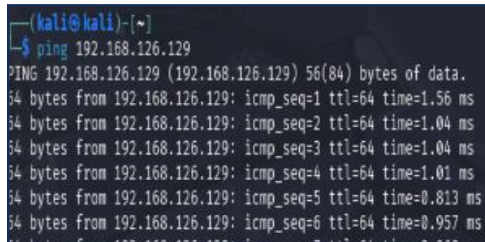
The first step was to verify connectivity between Kali Linux and Metasploitable2 using the ping command.

Command Used:

```
ping 192.168.126.129
```

Result:

The target machine responded successfully with 0% packet loss, confirming that communication between both systems was working correctly.



```
(kali@kali)-[~]  
$ ping 192.168.126.129  
PING 192.168.126.129 (192.168.126.129) 56(84) bytes of data.  
64 bytes from 192.168.126.129: icmp_seq=1 ttl=64 time=1.56 ms  
64 bytes from 192.168.126.129: icmp_seq=2 ttl=64 time=1.04 ms  
64 bytes from 192.168.126.129: icmp_seq=3 ttl=64 time=1.04 ms  
64 bytes from 192.168.126.129: icmp_seq=4 ttl=64 time=1.01 ms  
64 bytes from 192.168.126.129: icmp_seq=5 ttl=64 time=0.813 ms  
64 bytes from 192.168.126.129: icmp_seq=6 ttl=64 time=0.957 ms
```

6. Host Discovery Scan

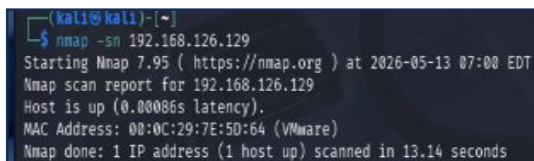
A host discovery scan was performed to verify that the target host was active on the network.

Command Used:

```
nmap -sn 192.168.126.129
```

Result:

The scan confirmed that the host was online and reachable.



```
(kali@kali)-[~]  
$ nmap -sn 192.168.126.129  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-13 07:00 EDT  
Nmap scan report for 192.168.126.129  
Host is up (0.00086s latency).  
MAC Address: 00:0C:29:7E:50:64 (VMware)  
Nmap done: 1 IP address (1 host up) scanned in 13.14 seconds
```

7. Basic Port Scan

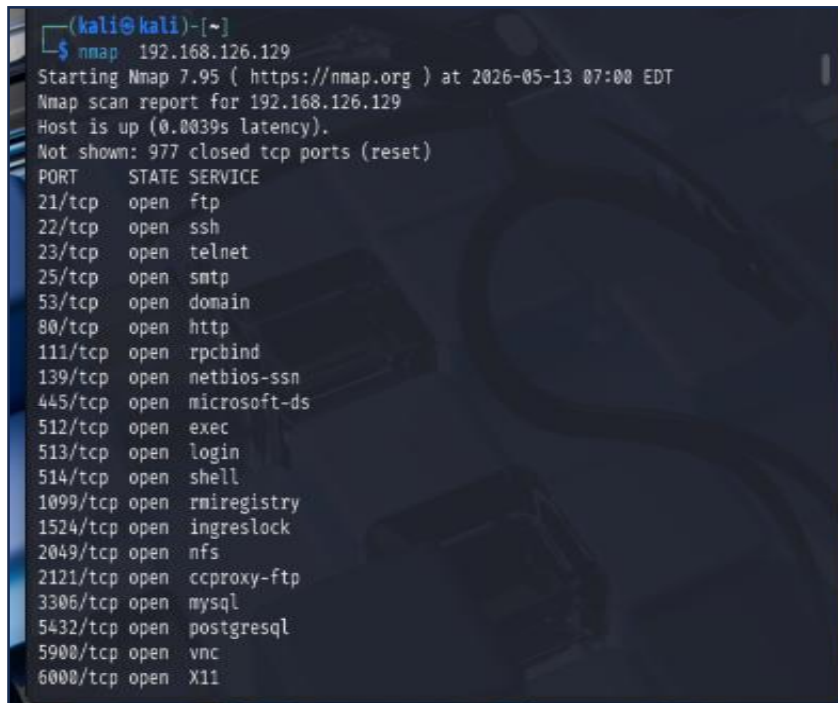
A basic Nmap scan was performed to identify open ports and active services.

Command Used

nmap 192.168.126.129

Result

Several ports were found open including FTP, SSH, Telnet, HTTP, SMB, MySQL, PostgreSQL, and Tomcat services.

A terminal window screenshot from a Kali Linux machine. The prompt is (kali@kali)-[~]. The command executed is nmap 192.168.126.129. The output shows the scan starting at 2026-05-13 07:00 EDT. It reports that the host is up with a latency of 0.0039s. It lists 977 closed TCP ports (not shown). A table of open ports follows, with columns for PORT, STATE, and SERVICE. The open ports are: 21/tcp (ftp), 22/tcp (ssh), 23/tcp (telnet), 25/tcp (smtp), 53/tcp (domain), 80/tcp (http), 111/tcp (rpcbind), 139/tcp (netbios-ssn), 445/tcp (microsoft-ds), 512/tcp (exec), 513/tcp (login), 514/tcp (shell), 1099/tcp (rmiregistry), 1524/tcp (ingreslock), 2049/tcp (nfs), 2121/tcp (ccproxy-ftp), 3306/tcp (mysql), 5432/tcp (postgresql), 5900/tcp (vnc), and 6000/tcp (X11).

```
(kali@kali)-[~]  
$ nmap 192.168.126.129  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-13 07:00 EDT  
Nmap scan report for 192.168.126.129  
Host is up (0.0039s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11
```

8. Service Version Detection

A service version scan was performed to identify the exact versions of running services.

Command Used:

nmap -sV 192.168.126.129

Result

Nmap successfully detected service versions running on the target machine.

```
L$ nmap -sV 192.168.126.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-13 07:04 EDT
Nmap scan report for 192.168.126.129
Host is up (0.0017s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
```

Detected Services

Port	Service	Version
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH 4.7p1
23	Telnet	Linux telnetd
25	SMTP	Postfix smtpd
53	DNS	ISC BIND 9.4.2
80	HTTP	Apache httpd 2.2.8
139	SMB	Samba smbd 3.x
445	SMB	Samba smbd 3.0.20
3306	MySQL	MySQL 5.0.51a
5432	PostgreSQL	PostgreSQL 8.3
8180	HTTP	Apache Tomcat/Coyote JSP engine 1.1

9. Operating System Detection

Operating system detection was performed to identify the target OS.

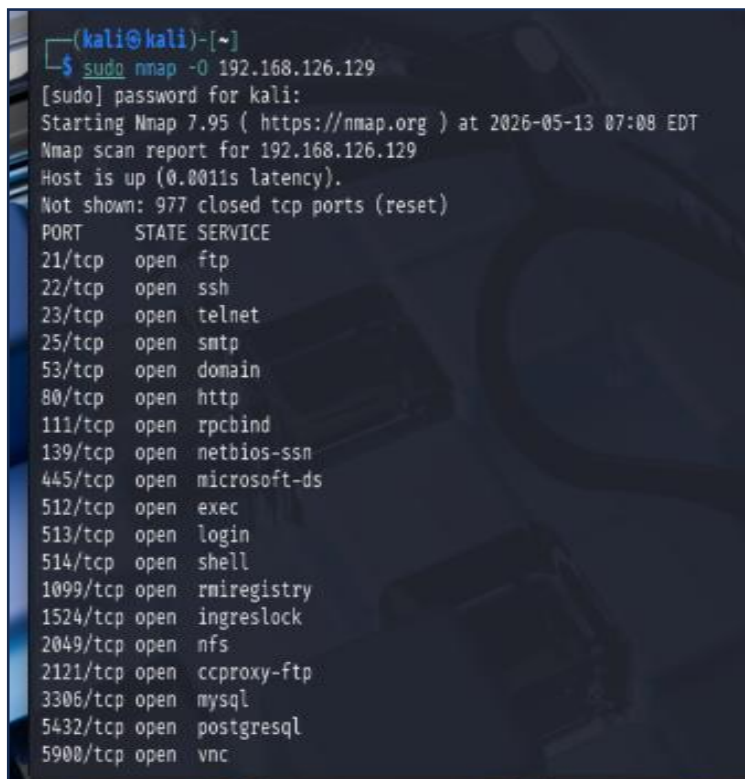
Command Used

```
sudo nmap -O 192.168.126.129
```

Result:

Nmap identified the operating system as:

Linux kernel 2.6.x

A terminal window screenshot from a Kali Linux machine. The prompt is (kali@kali)-[~]. The command executed is \$ sudo nmap -O 192.168.126.129. The output shows the Nmap scan report for 192.168.126.129, indicating the host is up with a latency of 0.0011s. It lists 20 open ports with their corresponding services. The background of the terminal has a dark blue theme with a faint Kali Linux logo.

```
(kali@kali)-[~]  
$ sudo nmap -O 192.168.126.129  
[sudo] password for kali:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-13 07:08 EDT  
Nmap scan report for 192.168.126.129  
Host is up (0.0011s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc
```

10. Aggressive Scan

An aggressive scan was performed to collect detailed information about the target machine.

Command Used

```
sudo nmap -A -O -sV -sR 192.168.126.129
```

Explanation of Parameters

Option	Meaning
-A	Aggressive scan
-O	Operating system detection
-sV	Service version detection
-sR	RPC scan
IP Address	Target machine

```

L$ nmap -A -O -sV -sR 192.168.126.129
WARNING: -sR is now an alias for -sV and activates version detection as well
as RPC scan.
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-13 08:12 EDT
Nmap scan report for 192.168.126.129
Host is up (0.00083s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 192.168.126.128
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   vsFTPD 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
```


11. Findings and Analysis

The scan revealed multiple outdated and vulnerable services running on the Metasploitable2 machine.

Important Findings

FTP Anonymous Login Enabled

The FTP service allowed anonymous login access.

Anonymous FTP login allowed

This represents a security vulnerability because unauthorized users may access files without authentication.

Samba Service Detected

SMB/Samba services were running on ports 139 and 445.

Samba smbd 3.0.20-Debian

Older Samba versions are known to contain vulnerabilities.

Apache Web Server

The Apache web server version detected was outdated.

Apache httpd 2.2.8

Outdated web servers may contain security flaws and known exploits.

Telnet Service Enabled

The Telnet service was active.

Linux telnetd

Telnet is insecure because communication is transmitted in plain text.

MySQL Database Service

The MySQL database service was accessible on port 3306.

MySQL 5.0.51a

12. Security Risks Identified

Vulnerability	Risk
Anonymous FTP Login	Unauthorized access
Telnet Enabled	Plain text communication
Old Samba Version	Remote exploitation
Outdated Apache Server	Web vulnerabilities
Multiple Open Ports	Increased attack surface

13. Conclusion

This practical demonstrated how Kali Linux and Nmap can be used to perform network reconnaissance and vulnerability assessment on a target system.

The scans successfully identified:

- Open ports
- Running services
- Service versions
- Operating system details
- Vulnerable configurations

Metasploitable2 intentionally contains insecure services, making it useful for cybersecurity learning and penetration testing practice.

This practical also helped improve understanding of:

- Network scanning
 - Enumeration
 - Service analysis
 - Vulnerability identification
-

1. Introduction

Nmap (Network Mapper) is an open-source network scanning and security auditing tool used to discover hosts, detect open ports, identify running services, perform operating system detection, and analyze network security.

Nmap is widely used by:

- Network administrators
- Cybersecurity analysts
- Penetration testers
- Ethical hackers

Official source: [Nmap Official Website](#)

2. Objectives

The objectives of this laboratory practical were to:

- Install and verify Nmap on Windows and Kali Linux
- Identify IP addresses of both systems
- Perform host discovery
- Perform port scanning
- Perform port selection scanning
- Detect services and versions

- Detect operating systems
- Use Nmap Scripting Engine (NSE)
- Perform aggressive professional scans
- Compare different Nmap scan techniques

3. Environment Used

Component	Description
Host Operating System	Windows 11
Virtual Machine	Kali Linux
Virtualization Software	VMware
Nmap Version (Windows)	7.99
Nmap Version (Kali Linux)	7.95
Target Machine	Kali Linux VM
Target IP Address	192.168.126.128
Open Service	Python HTTP Server
Open Port	8080

4. Verifying Nmap Installation

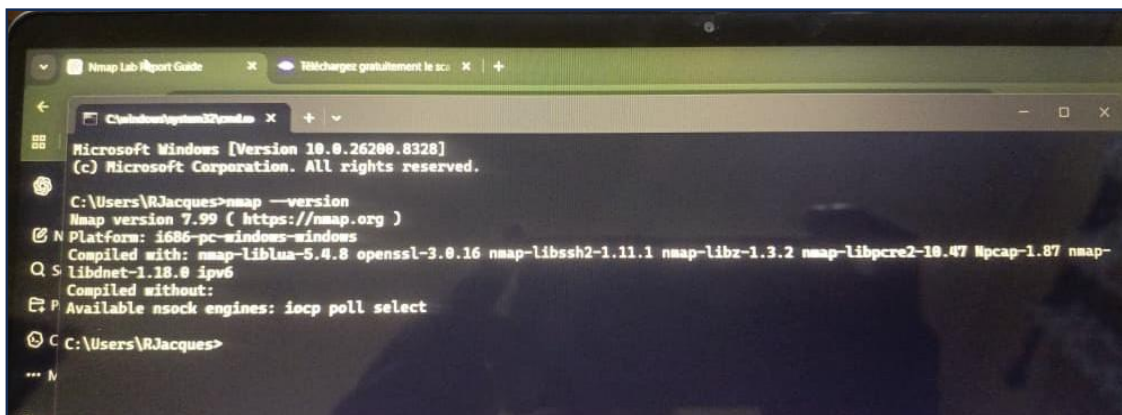
4.1. Windows Verification

Command:

```
nmap --version
```

Result:

Nmap version 7.99 was successfully installed on Windows.



```
Microsoft Windows [Version 10.0.26280.8328]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\RJacques>nmap --version  
Nmap version 7.99 ( https://nmap.org )  
Platform: i686-pc-windows-windows  
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 Npcap-1.87 nmap-  
libidnet-1.18.8 ipv6  
Compiled without:  
Available nsock engines: iocp poll select  
  
C:\Users\RJacques>
```

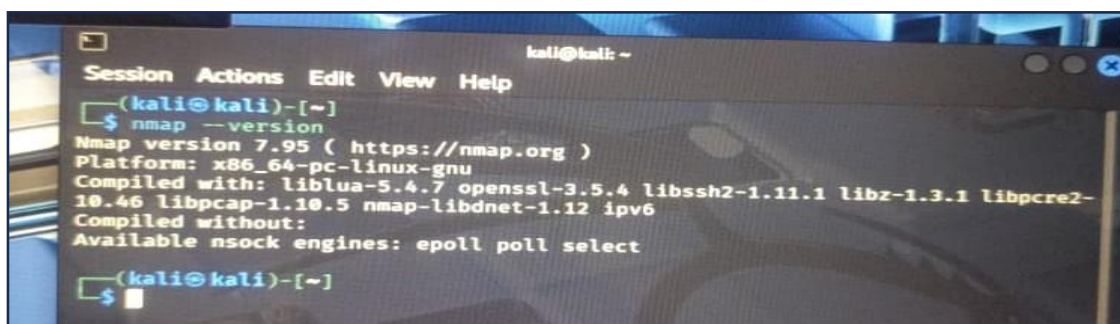
4.2. Kali Linux Verification

Command:

```
nmap --version
```

Result:

Nmap version 7.95 was successfully installed on Kali Linux.



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ nmap --version  
Nmap version 7.95 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.5 nmap-libidnet-1.12 ipv6  
Compiled without:  
Available nsock engines: epoll poll select  
  
(kali@kali)-[~]  
$
```

5. IP Address Configuration

5.1. Windows IP Address

Command:

Ipconfig

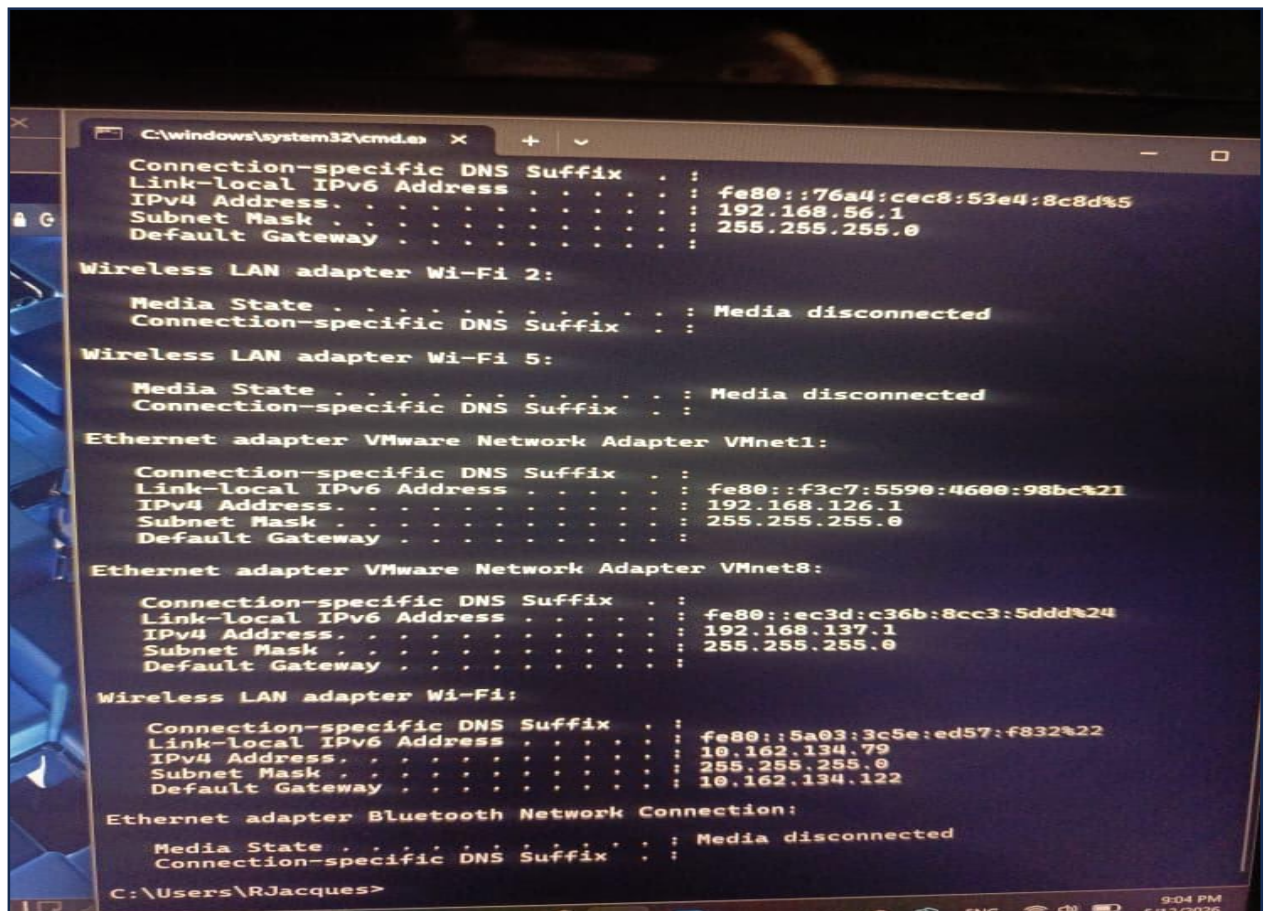
Result:

The Windows machine displayed the following network configuration:

Wi-Fi IPv4 Address: 10.162.134.79

VMnet1: 192.168.126.1

VMnet8: 192.168.137.1



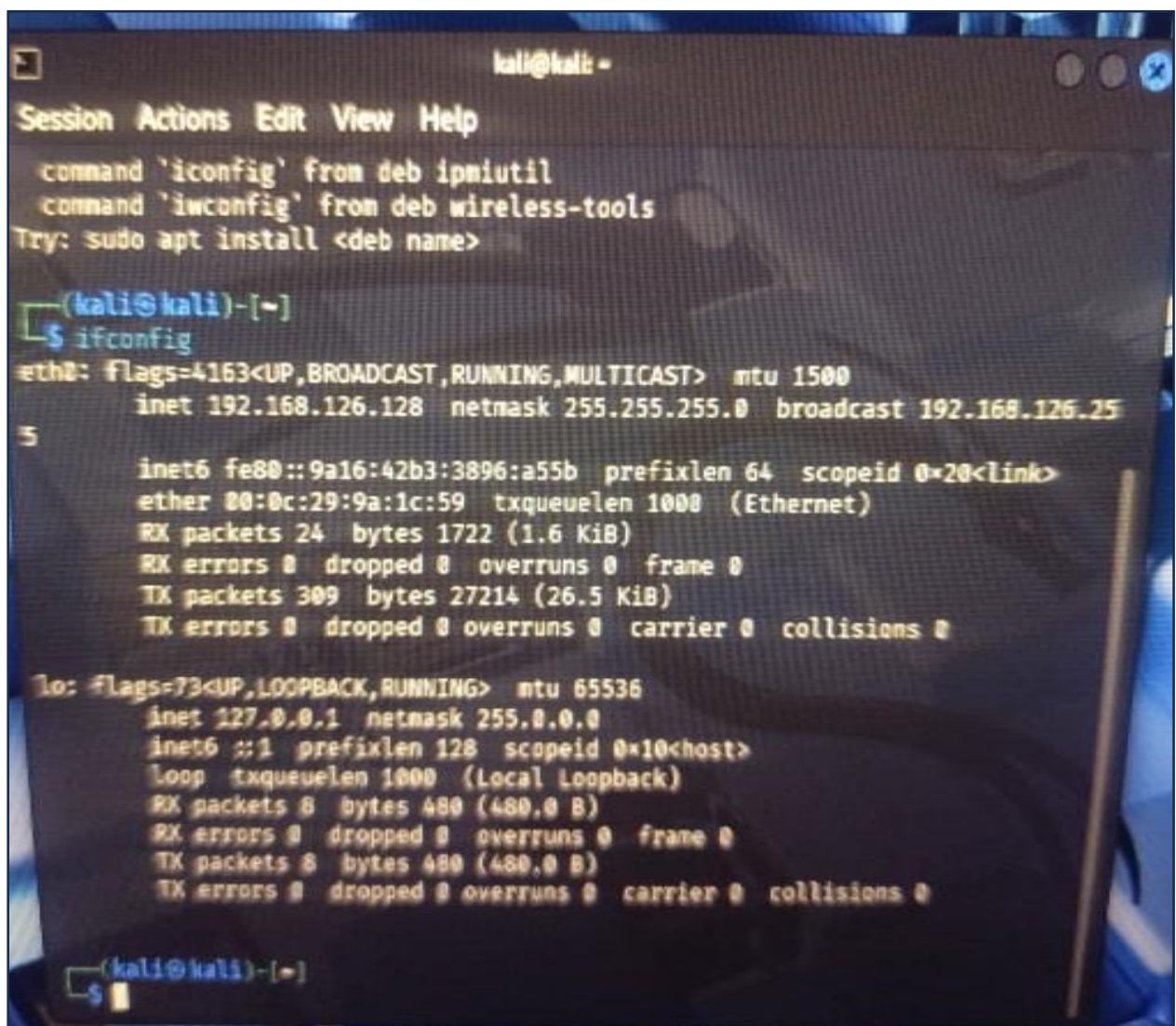
5.2. Kali Linux IP Address

Command:

Ifconfig

Result:

The Kali Linux virtual machine IP address was:
192.168.126.128



```
kali@kali: ~  
Session Actions Edit View Help  
command 'ifconfig' from deb ipmiutil  
command 'iwconfig' from deb wireless-tools  
Try: sudo apt install <deb name>  
  
(kali@kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.126.128 netmask 255.255.255.0 broadcast 192.168.126.255  
    inet6 fe80::9a16:42b3:3896:a55b prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:9a:1c:59 txqueuelen 1000 (Ethernet)  
    RX packets 24 bytes 1722 (1.6 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 309 bytes 27214 (26.5 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 8 bytes 480 (480.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 8 bytes 480 (480.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$
```


6. HOST DISCOVERY

Host discovery identifies whether a target machine is active on the network.

Command:

```
nmap -sn 192.168.126.128
```

Result:

Host is up

The target Kali Linux machine responded successfully, confirming that it was active and reachable.



7. INITIAL PORT SCANNING

Port scanning identifies open or closed ports on the target system.

Command:

```
nmap 192.168.126.128
```

Result:

All 1000 scanned ports are closed

Initially, no active services were running on the Kali Linux machine.

```
C:\Users\RJacques>nmap 192.168.126.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 21:13 +0000
Nmap scan report for 192.168.126.128
Host is up (0.00094s latency).
All 1000 scanned ports on 192.168.126.128 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:9A:1C:59 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.80 seconds
```

8. CREATING AN OPEN SERVICE

To demonstrate service detection, a Python HTTP server was started on Kali Linux.

Command:

```
python3 -m http.server 8080
```

Result:

Serving HTTP on 0.0.0.0 port 8080

The Python HTTP server created an active web service on port 8080.

```
(kali@kali)-[~]
$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

9. PORT SELECTION

Port selection scanning targets a specific port instead of scanning all ports.

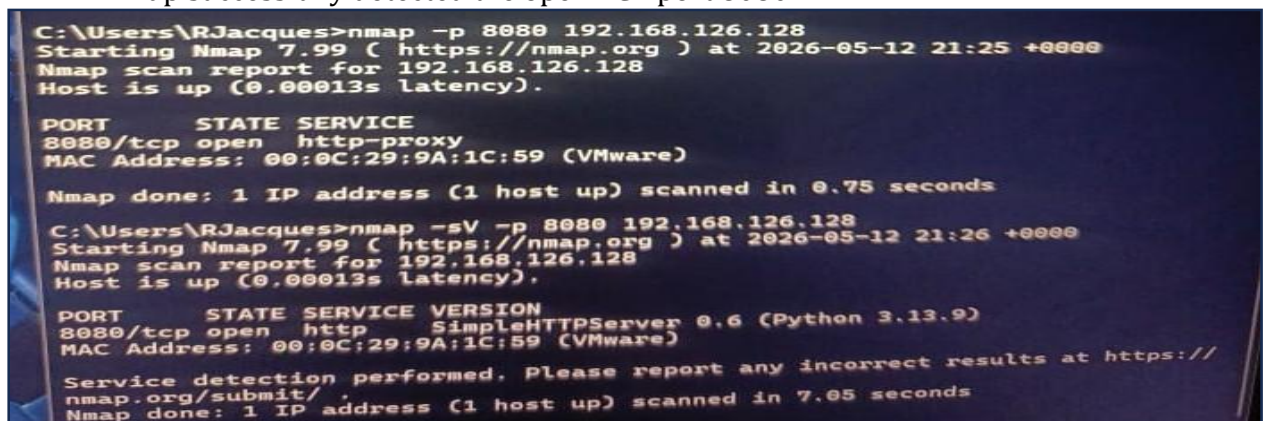
Command:

```
nmap -p 8080 192.168.126.128
```

Result:

8080/tcp open http-proxy

Nmap successfully detected the open TCP port 8080.



```
C:\Users\RJacques>nmap -p 8080 192.168.126.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 21:25 +0000
Nmap scan report for 192.168.126.128
Host is up (0.00013s latency).

PORT      STATE SERVICE
8080/tcp  open  http-proxy
MAC Address: 00:0C:29:9A:1C:59 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.75 seconds

C:\Users\RJacques>nmap -sV -p 8080 192.168.126.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 21:26 +0000
Nmap scan report for 192.168.126.128
Host is up (0.00013s latency).

PORT      STATE SERVICE VERSION
8080/tcp  open  http      SimpleHTTPServer 0.6 (Python 3.13.9)
MAC Address: 00:0C:29:9A:1C:59 (VMware)

Service detection performed. Please report any incorrect results at https://
nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.05 seconds
```

10. SERVICE VERSION DETECTION

Service version detection identifies the application and software version running on an open port.

Command:

```
nmap -sV 192.168.126.128
```

Result:

8080/tcp open http SimpleHTTPServer 0.6 (Python 3.13.9)

Nmap detected:

- the HTTP service,
- Python SimpleHTTPServer,

```
C:\Users\RJacques>nmap -sV 192.168.126.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 22:52 +0000
Nmap scan report for 192.168.126.128
Host is up (0.00014s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
8080/tcp  open  http      SimpleHTTPServer 0.6 (Python 3.13.9)
MAC Address: 00:0C:29:9A:1C:59 (VMware)

Service detection performed. Please report any incorrect results at https://
nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.04 seconds
```

11. NMAP SCRIPTING ENGINE (NSE)

NSE allows automated scripts to gather additional information from services.

Command:

```
nmap --script http-title -p 8080 192.168.126.128
```

Result:

http-title: Directory listing for /

The NSE script successfully retrieved the webpage title from the HTTP service.

```
C:\Users\RJacques>nmap --script http-title -p 8080 192.168.126.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 21:30 +0000
Nmap scan report for 192.168.126.128
Host is up (0.00013s latency).

PORT      STATE SERVICE
8080/tcp  open  http-proxy
|_http-title: Directory listing for /
MAC Address: 00:0C:29:9A:1C:59 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds
```

12. AGGRESSIVE SCANNING

Aggressive scanning combines multiple advanced scanning techniques.

Command:


```
nmap -A -O -sV 192.168.126.128
```

Features Performed

- Service detection
- Version detection
- OS detection
- NSE scripting
- Traceroute
- TCP/IP fingerprinting

Result:

SimpleHTTPServer 0.6 (Python 3.13.9)

Network Distance: 1 hop

Additional NSE information detected:

http-title: Directory listing for /

http-server-header: SimpleHTTP/0.6 Python/3.13.9

The aggressive scan provided much more detailed information than the normal service scan.

```
C:\Users\RJacques>nmap -A -O -sV 192.168.126.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 22:54 +0000
Nmap scan report for 192.168.126.128
Host is up (0.0010s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
8080/tcp  open  http      SimpleHTTPServer 0.6 (Python 3.13.9)
|_http-title: Directory listing for /
|_http-server-header: SimpleHTTP/0.6 Python/3.13.9
MAC Address: 08:0C:29:9A:1C:59 (VMware)
No exact OS matches for host (If you know what OS is running on it, see http
s://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=5/12%OT=8080%CT=1%CU=40455%PV=Y%DS=1%DC=D%G=Y%M=000C29
OS:%TM=6A03AFBE%P=i686-pc-windows-windows)SEQ(SP=101%GCD=1%ISR=10B%TI=Z%CI=
OS:Z%II=I%TS=A)SEQ(SP=105%GCD=1%ISR=10A%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=106%GCD=
OS:1%ISR=10B%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=106%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS
OS:=A)SEQ(SP=FF%GCD=1%ISR=103%TI=Z%CI=Z%II=I%TS=A)OPS(O1=M5B4ST11NW8%O2=M5B
OS:4ST11NW8%O3=M5B4NNT11NW8%O4=M5B4ST11NW8%O5=M5B4ST11NW8%O6=M5B4ST11)WIN(W
OS:1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%
OS:O=M5B4%CC=N%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R
OS:=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=
OS:AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=
OS:40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID
OS:=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

TRACEROUTE
HOP RTT ADDRESS
1 1.00 ms 192.168.126.128

OS and Service detection performed. Please report any incorrect results at h
ttps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.05 seconds
```

13. AGGRESSIVE SCAN WITH -sR

Command:

```
nmap -A -O -sV -sR 192.168.126.128
```

Result:

Nmap displayed the following warning:

WARNING: -sR is now an alias for -sV

This means:

- -sR is obsolete,
- RPC scanning is now integrated into -sV.

The scan still performed:

- OS detection,
- service detection,
- NSE scripting,
- traceroute.

```
C:\Users\RJacques>nmap -A -O -sV -sR 192.168.126.128
WARNING: -sR is now an alias for -sV and activates version detection as well
as RPC scan.
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-12 22:59 +0000
Nmap scan report for 192.168.126.128
Host is up (0.00089s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
8080/tcp  open  http    SimpleHTTPServer 0.6 (Python 3.13.9)
|_http-server-header: SimpleHTTP/0.6 Python/3.13.9
|_http-title: Directory listing for /
MAC Address: 00:0C:29:9A:1C:59 (VMware)
No exact OS matches for host (If you know what OS is running on it, see http
s://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=5/12%OT=8080%CT=1%CU=33160%PV=Y%DS=1%DC=D%G=Y%M=000C29
OS:%TM=6A03B0D0%P=i686-pc-windows-windows)SEQ(SP=100%GCD=1%ISR=108%TI=Z%CI=
OS:Z%II=I%TS=A)SEQ(SP=103%GCD=2%ISR=109%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=104%GCD=
OS:1%ISR=10A%TI=Z%CI=Z%II=I%TS=A)SEQ(SP=105%GCD=1%ISR=106%TI=Z%CI=Z%II=I%TS
OS:=A)SEQ(SP=FB%GCD=1%ISR=105%TI=Z%CI=Z%II=I%TS=A)OPS(O1=M5B4ST11NW8%O2=M5B
OS:4ST11NW8%O3=M5B4NNT11NW8%O4=M5B4ST11NW8%O5=M5B4ST11NW8%O6=M5B4ST11)WIN(W
OS:1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF8%
OS:O=M5B4%CC=N%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R
OS:Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=
OS:AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=
OS:40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID
OS:=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   0.89 ms  192.168.126.128

OS and Service detection performed. Please report any incorrect results at h
ttps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.13 seconds
```


14. HTTP SERVER LOG ANALYSIS

The Python HTTP server logs on Kali Linux confirmed that Nmap interacted with the web service.

Requests observed included:

- GET
- OPTIONS
- PROPFIND

Response codes observed:

- 200 OK
- 404 Not Found
- 501 Unsupported Method

These requests were generated automatically by Nmap during service fingerprinting and NSE analysis.

```
(kali@kali)-[~]
$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.126.1 - - [12/May/2026 17:26:39] "GET / HTTP/1.0" 200 -
192.168.126.1 - - [12/May/2026 17:26:39] code 501, message Unsupported method
('POST')
192.168.126.1 - - [12/May/2026 17:26:39] "POST /sdk HTTP/1.1" 501 -
192.168.126.1 - - [12/May/2026 17:26:39] "GET / HTTP/1.0" 200 -
192.168.126.1 - - [12/May/2026 17:26:39] code 404, message File not found
192.168.126.1 - - [12/May/2026 17:26:39] "GET /nmaplowercheck1778621199 HTTP/
1.1" 404 -
192.168.126.1 - - [12/May/2026 17:26:39] code 404, message File not found
192.168.126.1 - - [12/May/2026 17:26:39] "GET /HMAP1 HTTP/1.1" 404 -
192.168.126.1 - - [12/May/2026 17:26:39] code 404, message File not found
192.168.126.1 - - [12/May/2026 17:26:39] "GET /evox/about HTTP/1.1" 404 -
192.168.126.1 - - [12/May/2026 17:26:39] "GET / HTTP/1.0" 200 -
192.168.126.1 - - [12/May/2026 17:26:39] "GET / HTTP/1.1" 200 -
192.168.126.1 - - [12/May/2026 17:30:02] "GET / HTTP/1.1" 200 -
192.168.126.1 - - [12/May/2026 17:31:02] "GET / HTTP/1.0" 200 -
192.168.126.1 - - [12/May/2026 17:31:06] code 501, message Unsupported method
('OPTIONS')
192.168.126.1 - - [12/May/2026 17:31:06] "OPTIONS / HTTP/1.1" 501 -
192.168.126.1 - - [12/May/2026 17:31:06] code 501, message Unsupported method
('PROPFIND')
192.168.126.1 - - [12/May/2026 17:31:06] "PROPFIND / HTTP/1.1" 501 -
192.168.126.1 - - [12/May/2026 17:31:06] code 501, message Unsupported method
```

15. COMPARISON OF NMAP COMMANDS

Command	Purpose	Information Obtained
ifconfig	Display network information	IP addresses and interfaces
nmap -sV	Service version detection	Open ports and software versions
nmap -A -O -sV	Aggressive advanced scan	OS, services, NSE, traceroute
nmap -A -O -sV -sR	Old aggressive RPC scan	Same as above (-sR obsolete)

16. Security and Ethical Considerations

Nmap should only be used on:

- authorized systems,
- personal laboratory environments,
- networks with permission.

Unauthorized network scanning may violate:

- cybersecurity laws,
- company policies,
- internet service provider rules.

All experiments in this laboratory were performed in a controlled VMware virtual environment.

17. Conclusion

This practical laboratory successfully demonstrated the use of Nmap on both Windows and Kali Linux systems.

The experiments covered:

- Host discovery
- Port scanning
- Port selection
- Service detection
- Service version detection
- Operating system detection
- NSE scripting

- Aggressive scanning
- HTTP interaction analysis

The laboratory also demonstrated communication between a Windows host machine and a Kali Linux virtual machine using VMware networking.

Nmap proved to be a powerful and flexible tool for:

- network analysis,
- system discovery,
- cybersecurity auditing,
- penetration testing.

18. **References**

1. [Nmap Official Documentation](#)
2. [Nmap Reference Guide](#)
3. [Nmap NSE Documentation](#)
4. [Kali Linux Official Website](#)
5. [VMware Official Website](#)
6. [Metasploitable2 Project](#)
7. [VMware Official Website](#)